Linear transformations from F^n to F^m

Observation ①

Every element $(a_1, \ldots, a_n) = v \in F^n$

$$= \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Can be written uniquely in the form

$$v = a_1 e_1 + a_2 e_2 + \ldots + a_n e_n$$

where $e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow$ i-th position

i.e. $\{e_1, e_2, \ldots, e_n\}$ are a basis for F^n

Observation (2)

$$T : F^n \longrightarrow V$$ linear transformation

F-vector space

$$T(v) = T(a_1 e_1 + a_2 e_2 + \dots + a_n e_n)$$

$$= \underbrace{a_1 \cdot T(e_1) + a_2 \cdot T(e_2) + \dots + a_n \cdot T(e_n)}_{\in V}$$

i.e. T is determined completely by the n-vectors

$$v_1 = T(e_1), \dots, v_n = T(e_n) \in V$$

Observation (3)

Given $v_1, v_2, \dots, v_n \in V$, the rule

$$T : F^n \longrightarrow V$$

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \longmapsto \underbrace{a_1 v_1 + \dots + a_n v_n}$$

is a linear transformation

Pf:

(a) Verify that this gives a function:

- the entries of column vector determines it
- $a_1, \dots, a_n$ are the coeffs appearing when expressing the column vector uniquely as a linear combination of the standard basis vectors.

(b) Verify

$\cdot\ T\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right) = T\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right) + T\left(\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right)$

$c \in F$

$\cdot\ T\left(c \cdot \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right) = c \cdot T\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right)$

$$T\left(\begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}\right) = (a_1+b_1)v_1 + \dots + (a_n+b_n)v_n$$

$$= a_1v_1 + b_1v_1 + \dots + a_nv_n + b_nv_n$$

$\Vert$??

$$T\left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}\right) + T\left(\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}\right) = \begin{matrix} a_1v_1 + \dots + a_nv_n \\ + \\ b_1v_1 + \dots + b_nv_n \end{matrix}$$

Proposition: There is a bijection

$$\left\{ \begin{array}{l} \text{Linear transformations} \\ T : F^n \longrightarrow V \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} (v_1, \ldots, v_n) \\ \in \underbrace{V \times \cdots \times V}_{\text{n-fold product}} \end{array} \right\}$$

$$T \longmapsto (T(e_1), \ldots, T(e_n))$$

$$\underbrace{\left(\begin{array}{l} T : F^n \longrightarrow V \\ \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto c_1 v_1 + \cdots + c_n v_n \end{array} \right)} \longleftarrow\!\!| \; (\underbrace{v_1}, \ldots, \underbrace{v_n})$$

These functions are inverses to each other

Example:

(1) $T : F^3 \to \mathrm{Poly}(F)$

$$\rotatebox{90}{$\in$}$$

(a) $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \mapsto a_1 \cdot 1 + a_2 \cdot x + a_3 \cdot x^2$

(b) $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \mapsto a_1 \cdot x^{n_1} + a_2 \cdot x^{n_2} + a_3 \cdot x^{n_3}$

(c) Most generally, pick

$f_1(x), f_2(x), f_3(x) \in \mathrm{Poly}(F)$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \mapsto a_1 f_1(x) + a_2 f_2(x) + a_3 f_3(x)$$

will be a <u>linear transformation</u>

Corollary: $n, m \geq 1$

$$\left\{ \begin{matrix} T: F^n \longrightarrow F^m \\ \text{linear transformation} \end{matrix} \right\}$$

$$\updownarrow$$

$$\left\{ (v_1, \ldots, v_n) \in \underbrace{F^m \times \cdots \times F^m}_{n\text{-times}} \right\}$$

$$\updownarrow$$

$$\left\{ \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & & a_{m,n} \end{bmatrix} \in M_{m \times n}(F) \right\}$$

i.e. Giving a linear transformation

$$T : F^n \longrightarrow F^m$$

amounts to writing down an

$(m \times n)$-matrix $A \in M_{m \times n}(F)$

$$A = \underbrace{\begin{bmatrix} T(e_1) & T(e_2) & \cdots & T(e_n) \end{bmatrix}}_{\text{columns}} \Big\} \text{ m rows}$$

$$T(e_1), \ldots, T(e_n) \in F^m$$

Conversely, given:

$$A = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \Big\} \text{ m-rows}$$

$$v_1, \ldots, v_n \in F^m$$

we obtain

$$T : F^n \longrightarrow F^m$$

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \longmapsto a_1 v_1 + \cdots + a_n v_n \in F^m$$

If we take

$$e_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \\ \\ \\ \leftarrow i\text{-th} \\ \\ \\ \end{matrix}$$

$$T(e_i) = 0 \cdot v_1 + 0 \cdot v_2 + \cdots + 1 \cdot v_i + \cdots + 0 \cdot v_n$$

$$= v_i \in F^n$$

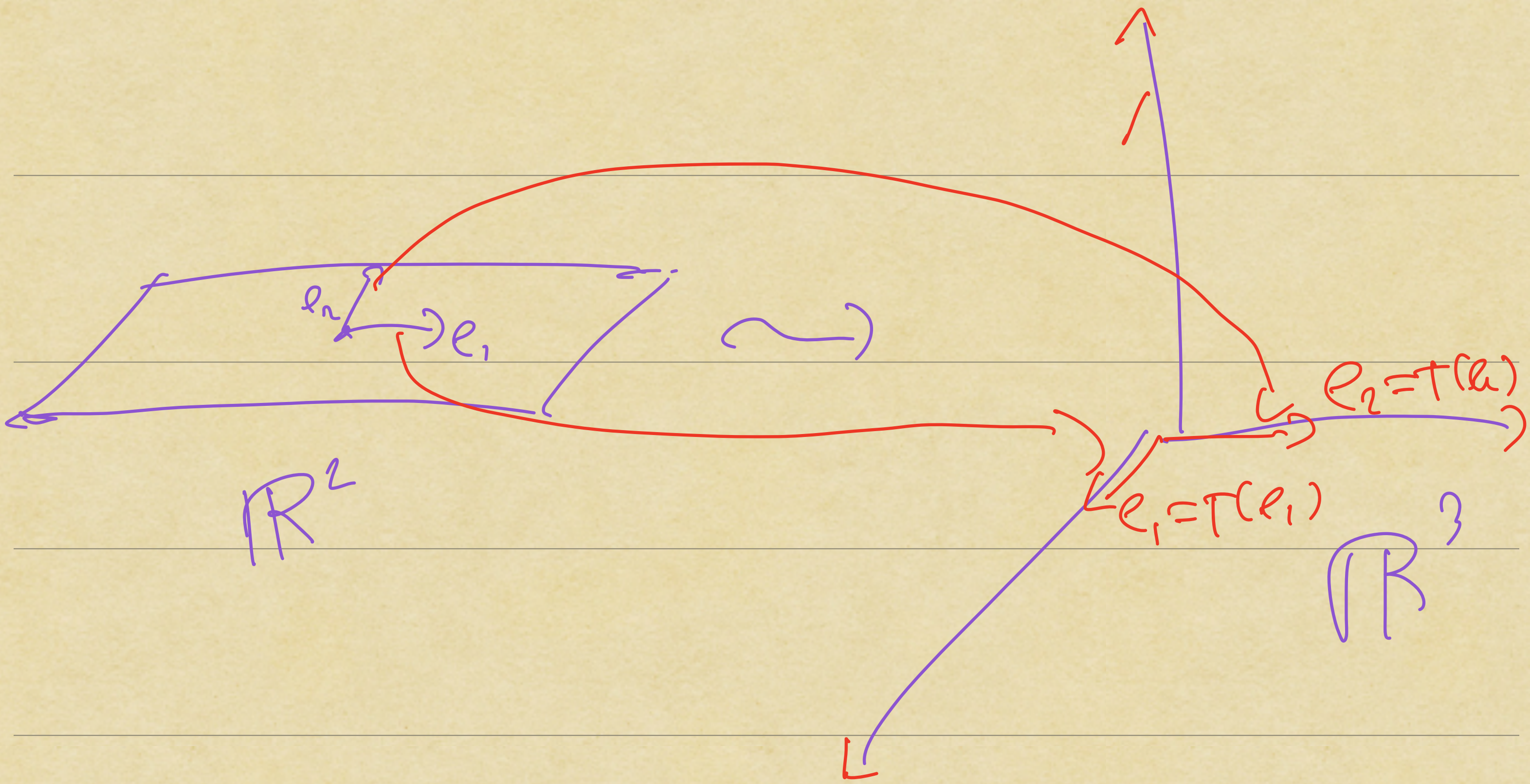
Example:

$$\{T : F^2 \longrightarrow \underline{F}^3\}$$

$$\longleftrightarrow \left\{ \begin{bmatrix} a & d \\ b & e \\ c & f \end{bmatrix} \in M_{3\times 2}(F) \right\}$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = T(e_1), \quad \begin{bmatrix} d \\ e \\ f \end{bmatrix} = T(e_2)$$

(a) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = T(e_1)$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = T(e_2)$



(b) $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = T(e_1)$, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = T(e_2)$